’Multi-Model Classification for CVD Prediction

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Set Working Directory

```
#setwd("C:/Users/ADMIN/OneDrive/Desktop/Eliud/R")
```

**Task 1 : Data Import and Exploration

Load required libraries

```
library(tidyverse)      # Data Manipulation
```

```
## ── Attaching core tidyverse packages ──────────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.0     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)          # Unified ML Interface
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(pROC)           # ROC Analysis
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```
library(e1071)          # SVM
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(class)          # KNN
library(randomForest)    # Random Forest
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following object is masked from 'package:randomForest':
##
##     outlier
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

Import the dataset

```
data <- read.csv("medical_data.csv")
head(data)
```

```
##        age gender      bmi blood_pressure cholesterol   glucose smoking alcohol
## 1 36.59287      0 19.88353       126.7126    207.0147 105.81544       1       0
## 2 41.54734      1 32.11152       125.8728    143.6313  76.89136       1       1
## 3 68.38062      1 30.43908       105.3434    156.2852 101.60278       1       0
## 4 46.05763      1 25.46273       140.1740    256.5752 130.69754       0       0
## 5 46.93932      0 26.90125       107.6709    138.6122  58.71636       0       0
## 6 70.72597      0 29.44199       135.1819    285.2456  78.22482       0       0
##   physical_activity family_history stress_level heart_rate sleep_hours
## 1          1.657959              0    3.3274065   77.77227    5.807189
## 2          1.552329              0    4.5588540   79.85721    5.297333
## 3          3.902421              0    0.7929705   77.91827    7.870152
## 4          3.492610              1    1.6643849   83.45057    7.777064
## 5          1.575045              0    2.8040743   90.25018    7.311528
## 6          5.070074              0    1.6687576   74.58525    6.274890
##   medication_adherence urban_residence CVD
## 1                    1               1  No
## 2                    0               1  No
## 3                    1               1  No
## 4                    1               0  No
## 5                    0               0  No
## 6                    1               1  No
```

**Explore the dataset

```
str(data)
```

```
## 'data.frame':    2500 obs. of  16 variables:
##  $ age                 : num  36.6 41.5 68.4 46.1 46.9 ...
##  $ gender              : int  0 1 1 1 0 0 0 1 1 0 ...
##  $ bmi                 : num  19.9 32.1 30.4 25.5 26.9 ...
##  $ blood_pressure      : num  127 126 105 140 108 ...
##  $ cholesterol         : num  207 144 156 257 139 ...
##  $ glucose             : num  105.8 76.9 101.6 130.7 58.7 ...
##  $ smoking             : int  1 1 1 0 0 0 0 0 0 0 ...
##  $ alcohol             : int  0 1 0 0 0 0 1 0 1 0 ...
##  $ physical_activity   : num  1.66 1.55 3.9 3.49 1.58 ...
##  $ family_history      : int  0 0 0 1 0 0 1 1 0 0 ...
##  $ stress_level        : num  3.327 4.559 0.793 1.664 2.804 ...
##  $ heart_rate          : num  77.8 79.9 77.9 83.5 90.3 ...
##  $ sleep_hours         : num  5.81 5.3 7.87 7.78 7.31 ...
##  $ medication_adherence: int  1 0 1 1 0 1 0 1 1 1 ...
##  $ urban_residence     : int  1 1 1 0 0 1 0 1 1 1 ...
##  $ CVD                 : chr  "No" "No" "No" "No" ...
```

```
describe(data)
```

```
##                      vars    n   mean    sd median trimmed   mad   min    max
## age                     1 2500  45.16 14.83  44.94   45.15 14.64 -0.72  95.86
## gender                  2 2500   0.49  0.50   0.00    0.49  0.00  0.00   1.00
## bmi                     3 2500  26.99  5.01  27.08   27.01  4.96 10.55  45.58
## blood_pressure          4 2500 130.09 15.05 129.92  130.08 15.65 72.32 187.72
## cholesterol             5 2500 199.80 40.02 199.01  199.72 39.38 63.92 322.70
## glucose                 6 2500  99.65 30.15  99.73   99.87 30.18  2.66 187.58
## smoking                 7 2500   0.29  0.45   0.00    0.24  0.00  0.00   1.00
## alcohol                 8 2500   0.38  0.49   0.00    0.36  0.00  0.00   1.00
## physical_activity       9 2500   2.93  1.49   2.91    2.92  1.47 -2.13   8.56
## family_history         10 2500   0.25  0.43   0.00    0.18  0.00  0.00   1.00
## stress_level           11 2500   4.97  2.01   4.99    4.98  1.97 -1.56  12.84
## heart_rate             12 2500  74.96 10.15  74.95   74.92 10.63 39.54 118.23
## sleep_hours            13 2500   6.95  1.49   6.92    6.95  1.51  2.54  12.25
## medication_adherence   14 2500   0.68  0.47   1.00    0.72  0.00  0.00   1.00
## urban_residence        15 2500   0.60  0.49   1.00    0.62  0.00  0.00   1.00
## CVD*                   16 2500   1.10  0.30   1.00    1.00  0.00  1.00   2.00
##                       range  skew kurtosis   se
## age                   96.57  0.03    -0.03 0.30
## gender                 1.00  0.03    -2.00 0.01
## bmi                   35.03 -0.03     0.07 0.10
## blood_pressure       115.40 -0.01     0.04 0.30
## cholesterol          258.78  0.03    -0.01 0.80
## glucose              184.92 -0.08    -0.06 0.60
## smoking                1.00  0.92    -1.15 0.01
## alcohol                1.00  0.48    -1.77 0.01
## physical_activity     10.68  0.10     0.03 0.03
## family_history         1.00  1.17    -0.63 0.01
## stress_level          14.40 -0.01    -0.01 0.04
## heart_rate            78.69  0.03    -0.06 0.20
## sleep_hours            9.71  0.04    -0.13 0.03
## medication_adherence   1.00 -0.77    -1.41 0.01
## urban_residence        1.00 -0.39    -1.85 0.01
## CVD*                   1.00  2.64     4.99 0.01
```

Check for missing values

```
# Count missing values per column
missing_count <- colSums(is.na(data))
missing_count
```

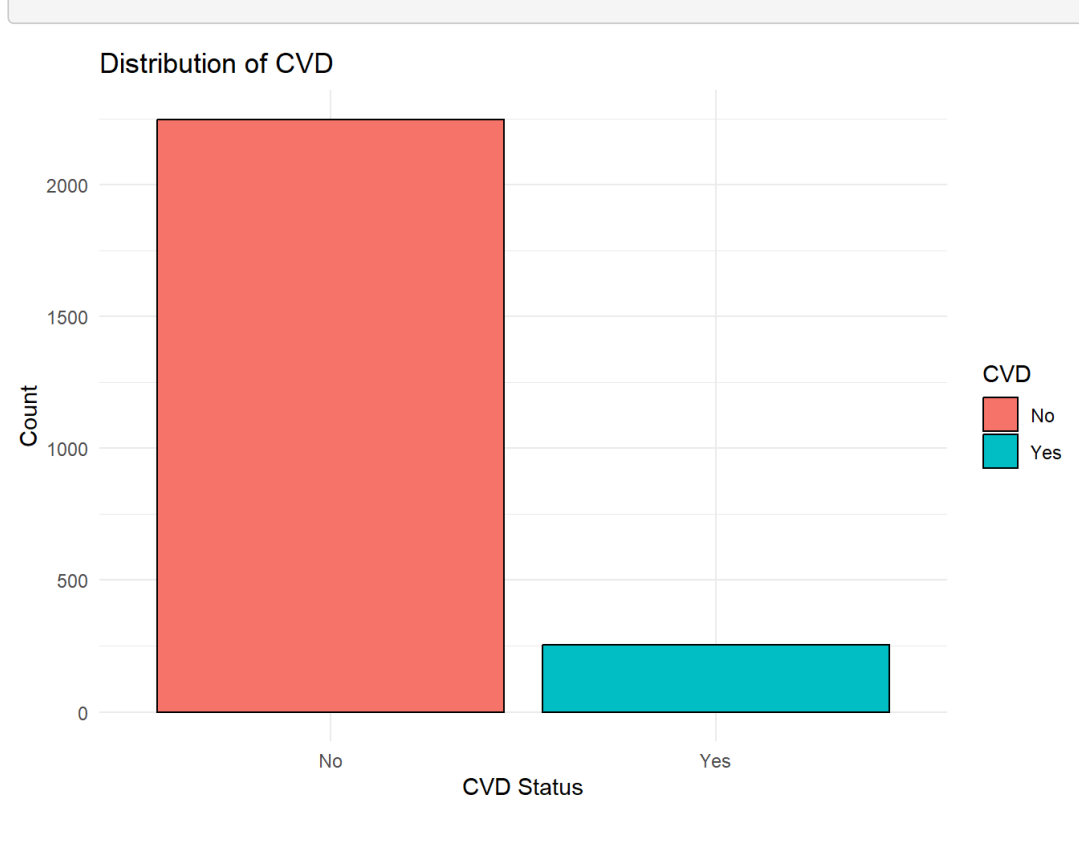
```
##                  age               gender                  bmi
##                    0                    0                    0
##       blood_pressure          cholesterol              glucose
##                    0                    0                    0
##              smoking              alcohol    physical_activity
##                    0                    0                    0
##       family_history         stress_level           heart_rate
##                    0                    0                    0
##          sleep_hours medication_adherence      urban_residence
##                    0                    0                    0
##                  CVD
##                    0
```

```
# Percentage of missing values
missing_percent <- (missing_count / nrow(data)) * 100
missing_percent
```

```
##                  age               gender                  bmi
##                    0                    0                    0
##       blood_pressure          cholesterol              glucose
##                    0                    0                    0
##              smoking              alcohol    physical_activity
##                    0                    0                    0
##       family_history         stress_level           heart_rate
##                    0                    0                    0
##          sleep_hours medication_adherence      urban_residence
##                    0                    0                    0
##                  CVD
##                    0
```

Visualize the distribution of the target variable CVD

```
ggplot(data, aes(x = CVD,fill = CVD)) +
  geom_bar(color = "black") +
  labs(title = "Distribution of CVD", x = "CVD Status", y = "Count") +
  theme_minimal()
```



Class Imbalance Ratio

```
# Count each class
class_counts <- table(data$CVD)

# Display counts
class_counts
```

```
##
##   No  Yes
## 2247  253
```

```
# Calculate ratio
imbalance_ratio <- max(class_counts) / min(class_counts)

imbalance_ratio
```

```
## [1] 8.881423
```

Task 2 : Data Preprocessing

Convert categorical variables to factors

```
data$gender <- as.factor(data$gender)
data$smoking <- as.factor(data$smoking)
data$alcohol <- as.factor(data$alcohol)
data$urban_residence <- as.factor(data$urban_residence)
data$family_history <- as.factor(data$family_history)
data$medication_adherence <- as.factor(data$medication_adherence)
```

Task 3 : Data Partitioning

Split the dataset into training and testing sets

```
set.seed(2026)
train_index <- createDataPartition(data$CVD, p = 0.7, list = FALSE)
train_data <- data[train_index, ]
test_data <- data[-train_index, ]
```

```
# Verify class distribution
cat("Training Set:\n")
```

```
## Training Set:
```

```
print(table(train_data$CVD))
```

```
##
##   No  Yes
## 1573  178
```

```
prop.table(table(train_data$CVD))
```

```
##
##        No       Yes
## 0.8983438 0.1016562
```

```
cat("\nTesting Set:\n")
```

```
##
## Testing Set:
```

```
print(table(test_data$CVD))
```

```
##
##  No Yes
## 674  75
```

```
prop.table(table(test_data$CVD))
```

```
##
##        No       Yes
## 0.8998665 0.1001335
```

```
preproc <- preProcess(train_data[, sapply(train_data, is.numeric)],
                      method = c("center", "scale"))

train_scaled <- predict(preproc, train_data)
test_scaled  <- predict(preproc, test_data)
```

Task 4 : Train Control Scheme

```
ctrl <- trainControl(
  method = "cv",               # 10-fold cross-validation
  number = 10,                # number of folds
  savePredictions = "final", # save predictions
  classProbs = TRUE,          # enable probabilities
  summaryFunction = twoClassSummary, # ROC, Sensitivity, Specificity
  verboseIter = FALSE         # suppress output
)
```

> Report :Cross-validation is used to evaluate model performance in a more reliable and robust manner. Instead of relying on a single train-test split, the data is divided into multiple folds, and the model is trained and validated across different subsets. This reduces the risk of overfitting and ensures that the model generalizes well to unseen data. The use of 10-fold cross-validation provides a good balance between computational efficiency and reliable performance estimation.

Task 5 :Model Training _ Logistic Regression

```
# Train Logistic Regression
log_model <- train(
  CVD ~.,
  data = train_data,
  method = "glm",
  family = binomial,
  trControl = ctrl
)
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.
```

```
# view results usingtab_model
library(sjPlot)
```

```
##
## Attaching package: 'sjPlot'
```

```
## The following object is masked from 'package:ggplot2':
##
##     set_theme
```

```
tab_model(log_model$finalModel,
          show.intercept = TRUE,
          show.se = TRUE,
          show.stat = TRUE,
          show.r2 = TRUE,
          show.ci = FALSE)
```

	Dependent variable			
Predictors	*Odds Ratios*	*std. Error*	*Statistic*	*p*
(Intercept)	0.00	0.00	-7.46	**<0.001**
age	1.04	0.01	5.83	**<0.001**
gender1	1.49	0.29	2.08	**0.038**
bmi	1.09	0.02	4.71	**<0.001**
blood pressure	1.02	0.01	3.44	**0.001**
cholesterol	1.02	0.00	5.97	**<0.001**
glucose	1.03	0.00	7.42	**<0.001**
smoking1	2.30	0.46	4.18	**<0.001**
alcohol1	1.26	0.24	1.18	0.236
physical activity	0.58	0.04	-7.50	**<0.001**
family history1	2.99	0.60	5.43	**<0.001**
stress level	1.40	0.07	6.66	**<0.001**
heart rate	1.00	0.01	-0.31	0.757
sleep hours	0.57	0.04	-7.83	**<0.001**
medication adherence1	0.47	0.09	-3.91	**<0.001**
urban residence1	0.93	0.18	-0.40	0.692
Observations	1751			
R² Tjur	0.299			

```
# Extract Coefficients
summary(log_model$finalModel)$coefficients
```

```
##                          Estimate  Std. Error    z value     Pr(>|z|)
## (Intercept)          -12.223388973 1.637617069 -7.4641314 8.385090e-14
## age                    0.038929475 0.006681531  5.8264308 5.662532e-09
## gender1                0.398504901 0.191982303  2.0757377 3.791822e-02
## bmi                    0.090247597 0.019163271  4.7094045 2.484416e-06
## blood_pressure         0.022018579 0.006396773  3.4421386 5.771345e-04
## cholesterol            0.015507712 0.002598662  5.9675759 2.408042e-09
## glucose                0.025243506 0.003401552  7.4211733 1.160873e-13
## smoking1               0.834518937 0.199868435  4.1753413 2.975396e-05
## alcohol1               0.228565180 0.192885094  1.1849810 2.360249e-01
## physical_activity     -0.546484798 0.072872645 -7.4991761 6.422021e-14
## family_history1        1.094374432 0.201720271  5.4252080 5.788708e-08
## stress_level           0.336797163 0.050543128  6.6635600 2.672729e-11
## heart_rate            -0.002834115 0.009157998 -0.3094688 7.569649e-01
## sleep_hours           -0.566593016 0.072403269 -7.8255171 5.055745e-15
## medication_adherence1 -0.762130752 0.195053682 -3.9072872 9.333819e-05
## urban_residence1      -0.075813946 0.191343530 -0.3962190 6.919435e-01
```

```
# Obtain Odds Ratios
coefficients <- coef(log_model$finalModel)
odds_ratios <- exp(coefficients)

round(odds_ratios, 3)
```

```
##          (Intercept)                  age              gender1
##                0.000                1.040                1.490
##                  bmi       blood_pressure          cholesterol
##                1.094                1.022                1.016
##              glucose             smoking1             alcohol1
##                1.026                2.304                1.257
##    physical_activity      family_history1         stress_level
##                0.579                2.987                1.400
##           heart_rate          sleep_hours medication_adherence1
##                0.997                0.567                0.467
##     urban_residence1
##                0.927
```

Task 6 : Model Training - KNN

```
# Train KNN using scaled data
knn_model <- train(
  CVD ~ .,
  data = train_scaled,
  method = "knn",
  trControl = ctrl,
  tuneGrid = expand.grid(k = seq(1, 21, by = 2))
)
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.
```

View the model summary

```
knn_model$results
```

```
##     k       ROC      Sens       Spec      ROCSD      SensSD     SpecSD
## 1   1 0.5834224 0.9243610 0.24248366 0.04387034 0.019030976 0.09140985
## 2   3 0.6751697 0.9745747 0.14607843 0.06420853 0.011993983 0.07617565
## 3   5 0.7104877 0.9847376 0.12418301 0.06304455 0.006857032 0.06535948
## 4   7 0.7425336 0.9923688 0.07385621 0.05784612 0.005016389 0.07721460
## 5   9 0.7630693 0.9968233 0.07941176 0.04752888 0.003348516 0.07774833
## 6  11 0.7714290 0.9981013 0.05686275 0.04038654 0.003057252 0.05473138
## 7  13 0.7846386 0.9980972 0.05686275 0.04694622 0.003063763 0.04805885
## 8  15 0.7854912 0.9993671 0.05718954 0.04896473 0.002001442 0.06794540
## 9  17 0.7928202 0.9980932 0.04542484 0.04473066 0.003070254 0.05973334
## 10 19 0.7999286 0.9980932 0.04542484 0.05077345 0.003070254 0.05973334
## 11 21 0.8019186 0.9987301 0.02843137 0.04843683 0.002677104 0.02999305
```

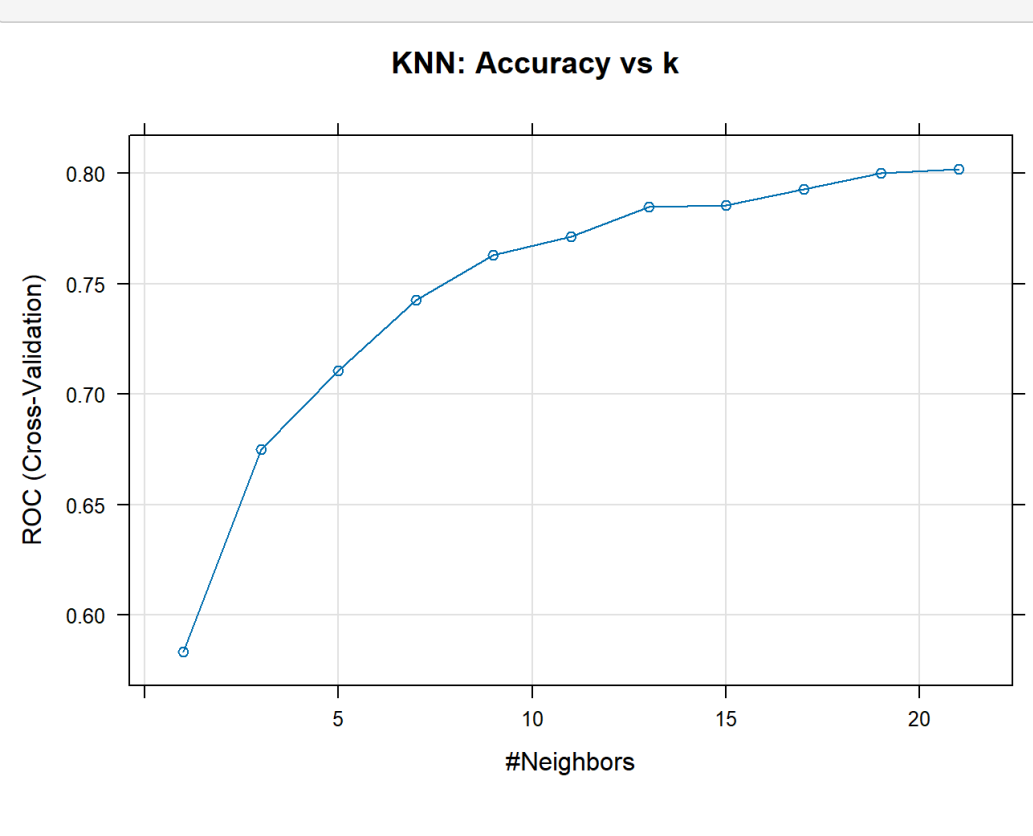
View the best tuned model

```
knn_model$bestTune
```

```
##     k
## 11 21
```

Plot Accuracy vs K

```
plot(knn_model, main = "KNN: Accuracy vs k")
```



Task 7 : Model Training -SVM

```
# Training SVM using scaled data
svm_model <- train(
  CVD ~.,
  data = train_scaled,
  method = "svmRadial",
  trControl = ctrl,
  tuneGrid = expand.grid(
    C = c(0.25, 0.5, 1, 2, 4),
    sigma = c(0.01, 0.05, 0.1, 0.5)
  )
)
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.
```

View the Model Summary

```
svm_model$results
```

```
##       C sigma       ROC      Sens        Spec      ROCSD      SensSD     SpecSD
## 1  0.25  0.01 0.8616624 0.9930098 0.198366013 0.04650497 0.005568919 0.16404411
## 5  0.50  0.01 0.8615748 0.9917399 0.197385621 0.04643147 0.006030851 0.14312250
## 9  1.00  0.01 0.8617591 0.9904660 0.207516340 0.04648603 0.008079462 0.11095347
## 13 2.00  0.01 0.8614691 0.9923688 0.151960784 0.04634244 0.005016389 0.09185597
## 17 4.00  0.01 0.8611894 0.9930098 0.186601307 0.04731614 0.004690138 0.14609504
## 2  0.25  0.05 0.8322277 0.9841047 0.253921569 0.06393898 0.006865449 0.12703646
## 6  0.50  0.05 0.8320471 0.9841047 0.259477124 0.06388886 0.006174053 0.13458516
## 10 1.00  0.05 0.8308396 0.9866565 0.243137255 0.06319628 0.006980447 0.14073973
## 14 2.00  0.05 0.8287168 0.9891841 0.208823529 0.06065842 0.006756353 0.11713608
## 18 4.00  0.05 0.8243674 0.9910949 0.169281046 0.06004024 0.006843648 0.10019181
## 3  0.25  0.10 0.8142769 0.9853826 0.203267974 0.07180940 0.007371633 0.12716483
## 7  0.50  0.10 0.8143810 0.9809280 0.231372549 0.07163490 0.006705743 0.13018956
## 11 1.00  0.10 0.8136690 0.9891881 0.180392157 0.07072791 0.007388505 0.10586980
## 15 2.00  0.10 0.8128572 0.9885471 0.151960784 0.06653439 0.008908404 0.08405818
## 19 4.00  0.10 0.8087416 0.9891800 0.123856209 0.05808528 0.009037153 0.08255112
## 4  0.25  0.50 0.7386284 0.9955414 0.022549020 0.05409068 0.006042569 0.03916876
## 8  0.50  0.50 0.7388127 0.9949045 0.016993464 0.05449617 0.007231162 0.02737653
## 12 1.00  0.50 0.7387383 0.9929936 0.016666667 0.05429781 0.008195439 0.03749714
## 16 2.00  0.50 0.7354990 0.9968153 0.005555556 0.05451212 0.006189970 0.01756821
## 20 4.00  0.50 0.7334459 0.9955414 0.005555556 0.05406085 0.006747452 0.01756821
```

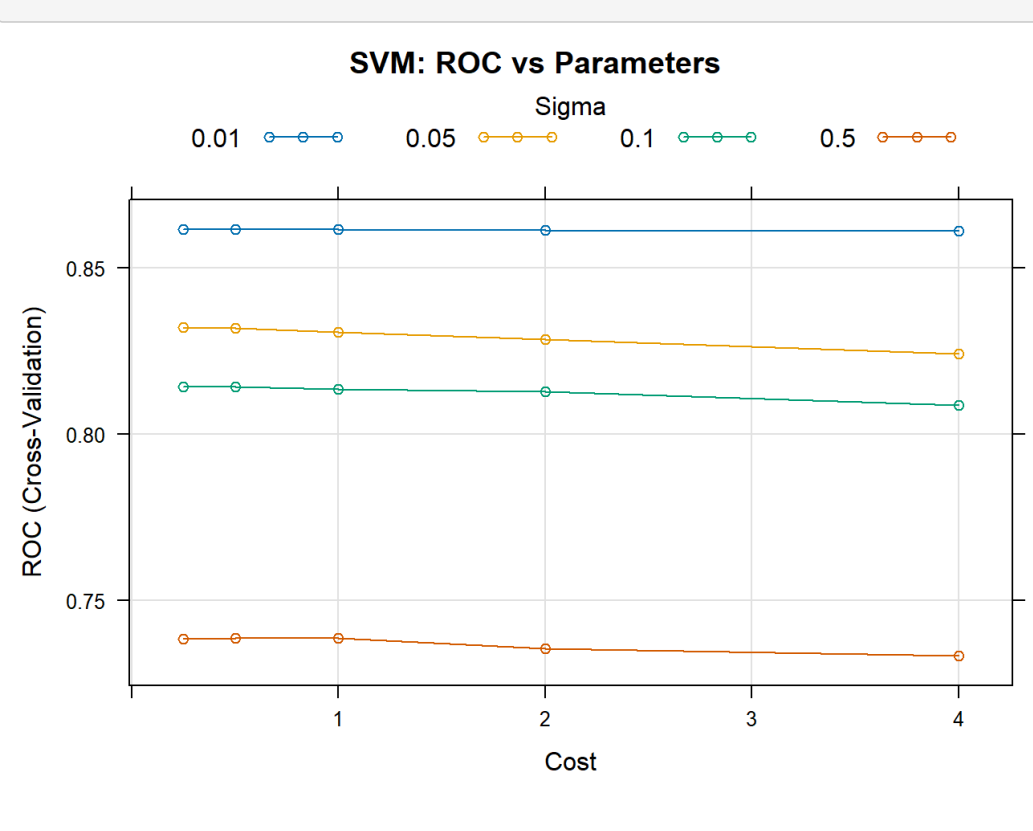
View Optimal Parameters

```
svm_model$bestTune
```

```
##   sigma C
## 9  0.01 1
```

Plot the Tuning Model

```
plot(svm_model, main = "SVM: ROC vs Parameters")
```



Task 8 : Model Training - Random Forest

```
rf_model <- train(
  CVD ~ .,
  data = train_data,
  method = "rf",
  trControl = ctrl,
  tuneGrid = expand.grid(mtry = c(2, 4, 6, 8)),
  ntree = 500
)
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.
```

View the Model Summary

```
rf_model$results
```

```
##   mtry       ROC      Sens        Spec      ROCSD      SensSD     SpecSD
## 1    2 0.8464018 0.9987301 0.005555556 0.03663613 0.002677104 0.01756821
## 2    4 0.8308202 0.9974563 0.062091503 0.04690759 0.003283974 0.04187883
## 3    6 0.8348427 0.9955495 0.079084967 0.04616979 0.005229672 0.03045437
## 4    8 0.8257209 0.9942796 0.096405229 0.04808469 0.004685872 0.05650330
```

View Optimal mtry

```
rf_model$bestTune
```

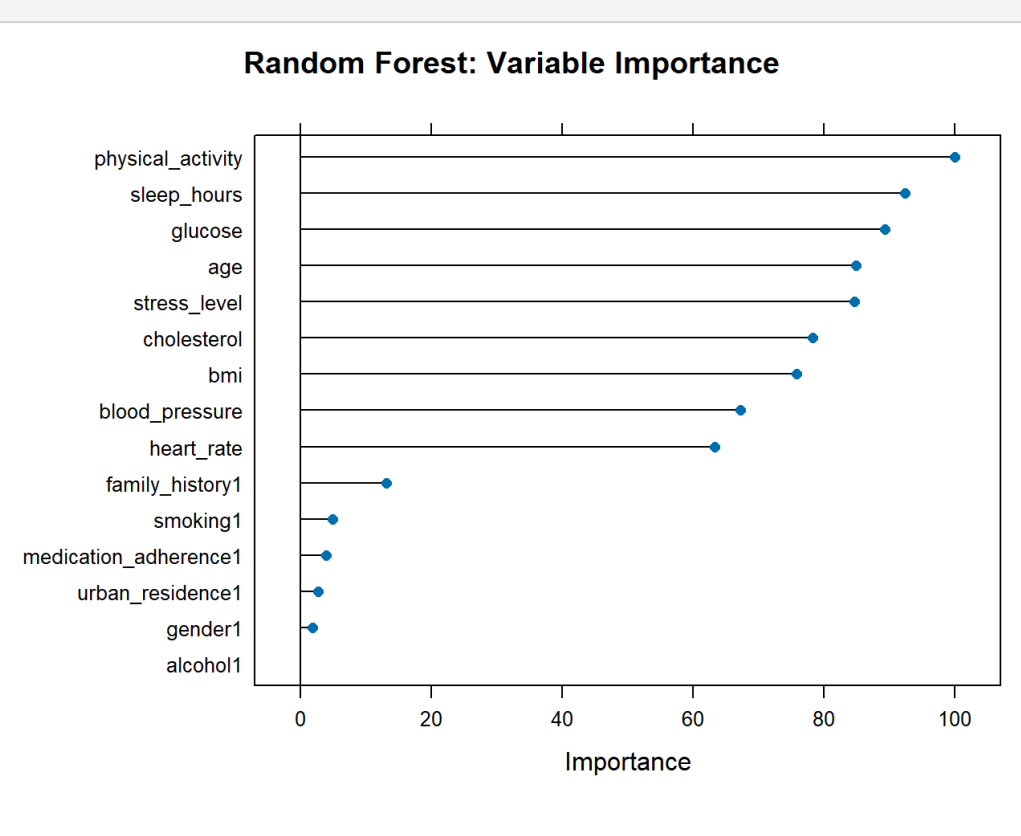
```
##   mtry
## 1    2
```

Variable Importance

```
importance <- varImp(rf_model)
print(importance)
```

```
## rf variable importance
##
##                       Overall
## physical_activity     100.000
## sleep_hours            92.342
## glucose                89.333
## age                    84.891
## stress_level           84.708
## cholesterol            78.325
## bmi                    75.844
## blood_pressure         67.188
## heart_rate             63.341
## family_history1        13.198
## smoking1                4.916
## medication_adherence1   3.895
## urban_residence1        2.695
## gender1                 1.887
## alcohol1                0.000
```

```
plot(importance, main = "Random Forest: Variable Importance")
```



Task 9 : Model Evaluation and Comparison

Predictions for all models

```
# Logistic Regression
log_pred  <- predict(log_model, test_data)
log_probs <- predict(log_model, test_data, type = "prob")

# KNN (scaled)
knn_pred  <- predict(knn_model, test_scaled)
knn_probs <- predict(knn_model, test_scaled, type = "prob")

# SVM (scaled)
svm_pred  <- predict(svm_model, test_scaled)
svm_probs <- predict(svm_model, test_scaled, type = "prob")

# Random Forest
rf_pred  <- predict(rf_model, test_data)
rf_probs <- predict(rf_model, test_data, type = "prob")
```

Confusion Matrices

```
# Convert test labels to factor with correct levels
test_data$CVD <- factor(test_data$CVD, levels = c("No", "Yes"))
test_scaled$CVD <- factor(test_scaled$CVD, levels = c("No", "Yes"))
```

```
# Logistic
cm_log <- confusionMatrix(log_pred, test_data$CVD, positive = "Yes")
cm_log
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  661  60
##        Yes  13  15
##
##                Accuracy : 0.9025
##                  95% CI : (0.879, 0.9228)
##     No Information Rate : 0.8999
##     P-Value [Acc > NIR] : 0.4338
##
##                   Kappa : 0.2505
##
##  Mcnemar's Test P-Value : 7.289e-08
##
##             Sensitivity : 0.20000
##             Specificity : 0.98071
##          Pos Pred Value : 0.53571
##          Neg Pred Value : 0.91678
##              Prevalence : 0.10013
##          Detection Rate : 0.02003
##    Detection Prevalence : 0.03738
##       Balanced Accuracy : 0.59036
##
##        'Positive' Class : Yes
##
```

```
# KNN
cm_knn <- confusionMatrix(knn_pred, test_scaled$CVD, positive = "Yes")
cm_knn
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  674  74
##        Yes   0   1
##
##                Accuracy : 0.9012
##                  95% CI : (0.8776, 0.9216)
##     No Information Rate : 0.8999
##     P-Value [Acc > NIR] : 0.4822
##
##                   Kappa : 0.0237
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.013333
##             Specificity : 1.000000
##          Pos Pred Value : 1.000000
##          Neg Pred Value : 0.901070
##              Prevalence : 0.100134
##          Detection Rate : 0.001335
##    Detection Prevalence : 0.001335
##       Balanced Accuracy : 0.506667
##
##        'Positive' Class : Yes
##
```

```
# SVM
cm_svm <- confusionMatrix(svm_pred, test_scaled$CVD, positive = "Yes")
cm_svm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  663  64
##        Yes  11  11
##
##                Accuracy : 0.8999
##                  95% CI : (0.8761, 0.9204)
##     No Information Rate : 0.8999
##     P-Value [Acc > NIR] : 0.5307
##
##                   Kappa : 0.19
##
##  Mcnemar's Test P-Value : 1.92e-09
##
##             Sensitivity : 0.14667
##             Specificity : 0.98368
##          Pos Pred Value : 0.50000
##          Neg Pred Value : 0.91197
##              Prevalence : 0.10013
##          Detection Rate : 0.01469
##    Detection Prevalence : 0.02937
##       Balanced Accuracy : 0.56517
##
##        'Positive' Class : Yes
##
```

```
# Random Forest
cm_rf <- confusionMatrix(rf_pred, test_data$CVD, positive = "Yes")
cm_rf
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  674  75
##        Yes   0   0
##
##                Accuracy : 0.8999
##                  95% CI : (0.8761, 0.9204)
##     No Information Rate : 0.8999
##     P-Value [Acc > NIR] : 0.5307
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.0000
##             Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.8999
##              Prevalence : 0.1001
##          Detection Rate : 0.0000
##    Detection Prevalence : 0.0000
##       Balanced Accuracy : 0.5000
##
##        'Positive' Class : Yes
##
```

ROC Curvess

```
log_roc <- roc(test_data$CVD, log_probs$Yes)
```

```
## Setting levels: control = No, case = Yes
```

```
## Setting direction: controls < cases
```

```
knn_roc <- roc(test_scaled$CVD, knn_probs$Yes)
```

```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

```
svm_roc <- roc(test_scaled$CVD, svm_probs$Yes)
```

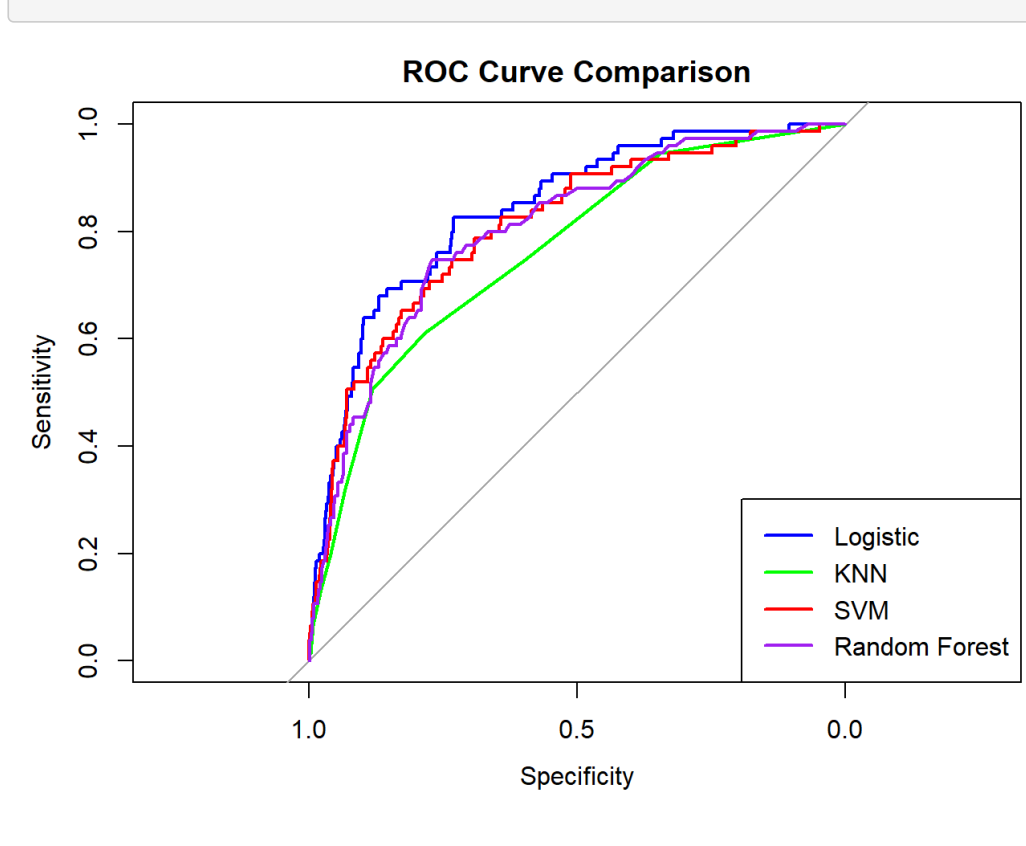
```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

```
rf_roc  <- roc(test_data$CVD, rf_probs$Yes)
```

```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

```
# Plot all
plot(log_roc, col = "blue", lwd = 2, main = "ROC Curve Comparison")
lines(knn_roc, col = "green", lwd = 2)
lines(svm_roc, col = "red", lwd = 2)
lines(rf_roc, col = "purple", lwd = 2)

legend("bottomright",
       legend = c("Logistic", "KNN", "SVM", "Random Forest"),
       col = c("blue", "green", "red", "purple"),
       lwd = 2)
```



AUC Comparison

```
auc_values <- data.frame(
  Model = c("Logistic", "KNN", "SVM", "Random Forest"),
  AUC = c(
    as.numeric(auc(log_roc)),
    as.numeric(auc(knn_roc)),
    as.numeric(auc(svm_roc)),
    as.numeric(auc(rf_roc))
  )
)

print(auc_values)
```

```
##           Model       AUC
## 1      Logistic 0.8395450
## 2           KNN 0.7626805
## 3           SVM 0.8083086
## 4 Random Forest 0.8035213
```